

All of Statistics: A Concise Course in Statistical Inference
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All errors, typographical and substantive, and other offenses, are entirely my own.

Chapter 5: Convergence of Random Variables

Question: 5.8.1

Let $X_1, \dots, X_n$ be IID with finite mean $\mu = \mathbb{E}(X_1)$ and finite variance $\sigma^2 = \mathbb{V}(X_1)$. Let $\bar{X}_n$ be the sample mean and let S_n^2 be the sample variance.

- (a) Show that $\mathbb{E}(S_n^2) = \sigma^2$.
- (b) Show that $S_n^2 \xrightarrow{P} \sigma^2$. Hint: Show that $S_n^2 = c_n n^{-1} \sum_{i=1}^n X_i^2 - d_n \bar{X}_n^2$ where $c_n \rightarrow 1$ and $d_n \rightarrow 1$. Apply the law of large numbers to $n^{-1} \sum_{i=1}^n X_i^2$ and to $\bar{X}_n$. Then use part (d) of Theorem 5.5.

PROOF. (a) See exercise 3.8.8, which provides a proof of Theorem 3.17.

(b) Using the hint, we can write S_n^2 as

$$\begin{aligned} S_n^2 &= \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \\ &= \frac{1}{n-1} \sum_{i=1}^n (X_i^2 - 2X_i \bar{X}_n + \bar{X}_n^2) \\ &= \frac{1}{n-1} \sum_{i=1}^n X_i^2 - \frac{n}{n-1} \bar{X}_n^2 \\ &= c_n \frac{1}{n} \sum_{i=1}^n X_i^2 - d_n \bar{X}_n^2 \end{aligned}$$

where $c_n = d_n = \frac{n}{n-1} \rightarrow 1$ as $n \rightarrow \infty$.

By WLLN, $\bar{X}_n \xrightarrow{P} \mu$. By Theorem 5.5(f), $\bar{X}_n^2 \xrightarrow{P} \mu^2$.

For the $\frac{1}{n} \sum_{i=1}^n X_i^2$ term, the task is to determine if $X_1^2, \dots, X_n^2$ are IID. Recall from exercise 2.14.10 that if $X \sqcup Y$, and if g, h are functions, then $g(X) \sqcup h(Y)$. Therefore, for X_i, X_j such that $i \neq j$, we must have $X_i^2 \sqcup X_j^2$. Since the X_i 's come from identical distributions, so must the X_i^2 's.

Thus, since $\mathbb{E}(X_1^2) = \sigma^2 + \mu^2$, by WLLN, $\frac{1}{n} \sum_{i=1}^n X_i^2 \xrightarrow{P} \sigma^2 + \mu^2$. Ergo, applying Theorem 5.5(d) to the products $c_n \frac{1}{n} \sum_{i=1}^n X_i^2$ and $d_n \bar{X}_n^2$, we find

$$S_n^2 = c_n \frac{1}{n} \sum_{i=1}^n X_i^2 - d_n \bar{X}_n^2 \xrightarrow{P} \sigma^2 + \mu^2 - \mu^2 = \sigma^2$$

□

Question: 5.8.2

Let $X_1, X_2, \dots$ be a sequence of random variables. Show that $X_n \xrightarrow{\text{qm}} b$ if and only if

$$\lim_{n \rightarrow \infty} \mathbb{E}(X_n) = b \quad \text{and} \quad \lim_{n \rightarrow \infty} \mathbb{V}(X_n) = 0.$$

PROOF. ($\implies$) Let $X_n \xrightarrow{\text{qm}} b$. Then $\mathbb{E}(X_n - b)^2 \rightarrow 0$. By Theorem 5.17(b), we also have that X_n converges in L_1 to b :

$$\mathbb{E}|X_n - b| \rightarrow 0$$

If $X_n - b \geq 0$, then $\mathbb{E}|X_n - b| = \mathbb{E}(X_n - b)$, and

$$\lim_{n \rightarrow \infty} \mathbb{E}(X_n - b) = \lim_{n \rightarrow \infty} \mathbb{E}(X_n) - b = 0 \implies \lim_{n \rightarrow \infty} \mathbb{E}(X_n) = b$$

Otherwise, $\mathbb{E}|X_n - b| = \mathbb{E}(-(X_n - b))$, and

$$\lim_{n \rightarrow \infty} \mathbb{E}(-(X_n - b)) = 0 \implies \lim_{n \rightarrow \infty} \mathbb{E}(X_n - b) \implies \lim_{n \rightarrow \infty} \mathbb{E}(X_n) = b$$

as before.

Thus we must have $\lim_{n \rightarrow \infty} \mathbb{E}(X_n) = b$ in either case. Combining this fact and the premise, it must follow that

$$\begin{aligned} \lim_{n \rightarrow \infty} \mathbb{E}(X_n - b)^2 &= \lim_{n \rightarrow \infty} \mathbb{E}(X_n^2 - 2bX_n + b^2) \\ &= \lim_{n \rightarrow \infty} \mathbb{E}(X_n^2) - 2b \lim_{n \rightarrow \infty} \mathbb{E}(X_n) + b^2 \\ &= \lim_{n \rightarrow \infty} \mathbb{E}(X_n^2) - 2 \lim_{n \rightarrow \infty} \mathbb{E}(X_n)^2 + \lim_{n \rightarrow \infty} \mathbb{E}(X_n)^2 \\ &= \lim_{n \rightarrow \infty} [\mathbb{E}(X_n^2) - \mathbb{E}(X_n)^2] \\ &= \lim_{n \rightarrow \infty} \mathbb{E}(X_n - \mathbb{E}(X_n))^2 \\ &= \lim_{n \rightarrow \infty} \mathbb{V}(X_n) \\ &= 0 \end{aligned}$$

($\impliedby$) In the converse direction, let

$$\lim_{n \rightarrow \infty} \mathbb{E}(X_n) = b \quad \lim_{n \rightarrow \infty} \mathbb{V}(X_n) = 0$$

Then

$$\begin{aligned} \lim_{n \rightarrow \infty} \mathbb{V}(X_n) &= \lim_{n \rightarrow \infty} \mathbb{E}(X_n - \mathbb{E}(X_n))^2 \\ &= \lim_{n \rightarrow \infty} \mathbb{E}(X_n^2 - 2X_n\mathbb{E}(X_n) + \mathbb{E}(X_n)^2) \\ &= \lim_{n \rightarrow \infty} \mathbb{E}(X_n^2) - 2 \lim_{n \rightarrow \infty} \mathbb{E}(X_n)^2 + \lim_{n \rightarrow \infty} \mathbb{E}(X_n)^2 \\ &= \lim_{n \rightarrow \infty} \mathbb{E}(X_n^2) - b^2 \\ &= \lim_{n \rightarrow \infty} \mathbb{E}(X_n - b)^2 \\ &= 0 \end{aligned}$$

implies $X_n \xrightarrow{\text{qm}} b$. □

Question: 5.8.3

Let $X_1, \dots, X_n$ be IID and let $\mu = \mathbb{E}(X_1)$. Suppose that the variance is finite. Show that $\bar{X}_n \xrightarrow{\text{qm}} \mu$.

PROOF. Since $\mathbb{E}(\bar{X}_n) = \mu$ and $\mathbb{V}(\bar{X}_n) = \sigma^2/n$, observe that

$$\begin{aligned}\lim_{n \rightarrow \infty} \mathbb{E}(\bar{X}_n) &= \lim_{n \rightarrow \infty} \mu = \mu \\ \lim_{n \rightarrow \infty} \mathbb{V}(\bar{X}_n) &= \lim_{n \rightarrow \infty} \frac{\sigma^2}{n} = 0\end{aligned}$$

By exercise 5.8.2, $\bar{X}_n \xrightarrow{\text{qm}} \mu$. □

Question: 5.8.4

Let $X_1, X_2, \dots$ be a sequence of random variables such that

$$\mathbb{P}\left(X_n = \frac{1}{n}\right) = 1 - \frac{1}{n^2} \quad \text{and} \quad \mathbb{P}(X_n = n) = \frac{1}{n^2}.$$

Does X_n converge in probability? Does X_n converge in quadratic mean?

By the errata, note that $n \geq 2$.

Conjecture: $X_n \xrightarrow{\text{P}} 0$

PROOF. Observe that $\lim_{n \rightarrow \infty} \mathbb{P}(|X_n| > \epsilon) = \mathbb{P}(n > \epsilon \text{ or } \frac{1}{n} > \epsilon)$. Then by mutual exclusivity:

$$\begin{aligned}&= \lim_{n \rightarrow \infty} \mathbb{P}(n > \epsilon) \mathbb{P}(X_n = n) + \lim_{n \rightarrow \infty} \mathbb{P}\left(\frac{1}{n} > \epsilon\right) \mathbb{P}\left(X_n = \frac{1}{n}\right) \\&= \lim_{n \rightarrow \infty} \mathbb{P}(n > \epsilon) \frac{1}{n^2} + \lim_{n \rightarrow \infty} \mathbb{P}\left(\frac{1}{n} > \epsilon\right) \left(1 - \frac{1}{n^2}\right) \\&= \lim_{n \rightarrow \infty} \mathbb{P}\left(\frac{1}{n} > \epsilon\right) \\&= 0\end{aligned}$$

Thus $X_n \xrightarrow{\text{P}} 0$. □

Conjecture: $X_n \xrightarrow{\text{qm}} X$

PROOF. Suppose X_n converges in quadratic mean to X . Then we have

$$\begin{aligned}\mathbb{E}(X_n - X)^2 &= \mathbb{E}(X_n^2 - 2X_nX + X^2) \\ &= \mathbb{E}(X_n^2) - 2\mathbb{E}(X_nX) + \mathbb{E}(X^2) \\ &= \frac{1}{n^2} \left(1 - \frac{1}{n^2}\right) + n^2 \left(\frac{1}{n^2}\right) \\ &\quad - 2 \left(\frac{1}{n} \mathbb{E}(X) \left(1 - \frac{1}{n^2}\right) + n \mathbb{E}(X) \left(\frac{1}{n^2}\right)\right) + \mathbb{E}(X^2)\end{aligned}$$

Now take the limit for large n to find

$$\lim_{n \rightarrow \infty} \mathbb{E}(X_n - X)^2 = 1 + \mathbb{E}(X^2)$$

Now, by our initial assumption, we must have

$$1 + \mathbb{E}(X^2) = 0$$

But this is impossible, as it would require $\mathbb{E}(X^2) = -1$, which can never be the case since $\mathbb{V}(X) \geq 0$. So $X_n \not\overset{\text{qm}}{\rightarrow} X$ for any X . $\square$

Question: 5.8.5

Let $X_1, \dots, X_n \sim \text{Bernoulli}(p)$. Prove that

$$\frac{1}{n} \sum_{i=1}^n X_i^2 \xrightarrow{P} p \quad \text{and} \quad \frac{1}{n} \sum_{i=1}^n X_i^2 \xrightarrow{\text{qm}} p.$$

PROOF. First we prove that $\frac{1}{n} \sum_{i=1}^n X_i^2 \xrightarrow{\text{qm}} p$. The expectation and mean are

$$\begin{aligned}\mathbb{E}\left(\frac{1}{n} \sum_{i=1}^n X_i^2\right) &= \frac{1}{n} \sum_{i=1}^n \mathbb{E}(X_i^2) \\ &= \frac{1}{n} np \\ &= p \\ \mathbb{V}\left(\frac{1}{n} \sum_{i=1}^n X_i^2\right) &= \frac{1}{n^2} \sum_{i=1}^n \mathbb{V}(X_i^2) \\ &= \frac{1}{n} \left[\mathbb{E}(X_i^4) - \mathbb{E}(X_i^2)^2\right] \\ &= \frac{1}{n} (p - p^2)\end{aligned}$$

So we ascertain that

$$\mathbb{E}\left(\frac{1}{n} \sum_{i=1}^n X_i^2 - p\right)^2 = \mathbb{V}\left(\frac{1}{n} \sum_{i=1}^n X_i^2\right)$$

and that the remaining task is to prove the variance vanishes in the limit:

$$\lim_{n \rightarrow \infty} \mathbb{V} \left(\frac{1}{n} \sum_{i=1}^n X_i^2 \right) = \lim_{n \rightarrow \infty} \frac{1}{n} (p - p^2) = 0$$

Thus $\frac{1}{n} \sum_{i=1}^n X_i^2 \xrightarrow{\text{qm}} p$. By Theorem 5.4(a), it immediately follows that $\frac{1}{n} \sum_{i=1}^n X_i^2 \xrightarrow{\text{P}} p$. $\square$

Question: 5.8.6

Suppose that the height of men has mean 68 inches and standard deviation 2.6 inches. We draw 100 men at random. Find (approximately) the probability that the average height of men in our sample will be at least 72 inches.

Note that the errata is for 72 inches.

For $Z \sim N(0, 1)$, by the Central Limit Theorem:

$$\begin{aligned} \mathbb{P}(\bar{X}_n > 72) &= \mathbb{P}\left(\frac{\sqrt{100}(\bar{X}_n - 68)}{2.6} > \frac{\sqrt{100}(72 - 68)}{2.6}\right) \\ &\approx \mathbb{P}(10Z > 15.38) \\ &= \mathbb{P}(Z > 1.538) \\ &\approx \boxed{0.062} \end{aligned}$$

Question: 5.8.7

Let $\lambda_n = 1/n$ for $n = 1, 2, \dots$. Let $X_n \sim \text{Poisson}(\lambda_n)$.

- (a) Show that $X_n \xrightarrow{\text{P}} 0$.
- (b) Let $Y_n = nX_n$. Show that $Y_n \xrightarrow{\text{P}} 0$.

PROOF. (a) Observe that

$$\lim_{n \rightarrow \infty} \mathbb{E}(X_n^2) = \lim_{n \rightarrow \infty} \mathbb{V}(X_n) + \mathbb{E}(X_n)^2 = \lim_{n \rightarrow \infty} \frac{1}{n} + \frac{1}{n^2} = 0$$

which implies $X_n \xrightarrow{\text{qm}} 0$, and by Theorem 5.4(a), we have $X_n \xrightarrow{\text{P}} 0$.

(b) By Theorem 5.5(f), if $Y_n = g(X_n) = nX_n$, then $X_n \xrightarrow{\text{P}} 0 \implies g(X_n) \xrightarrow{\text{P}} g(0) \implies Y_n \xrightarrow{\text{P}} 0$. $\square$

Question: 5.8.8

Suppose we have a computer program consisting of $n = 100$ pages of code. Let X_i be the number of errors on the i^{th} page of code. Suppose that the X_i 's are Poisson with mean 1 and that they are independent. Let $Y = \sum_{i=1}^n X_i$ be the total number of errors. Use the central limit theorem to approximate $\mathbb{P}(Y < 90)$.

By the Central Limit Theorem, $n\bar{X}_n \approx N(n\mu, n\sigma^2)$. Then calculate

$$\mathbb{P}(Y < 90) = \mathbb{P}\left(\frac{n\bar{X}_n - n\mu}{\sqrt{n\sigma^2}} < \frac{90 - (100)(1)}{\sqrt{100 \cdot 1}}\right) = \mathbb{P}(Z < -1) \approx \boxed{0.1587}$$

Question: 5.8.9

Suppose that $\mathbb{P}(X = 1) = \mathbb{P}(X = -1) = 1/2$. Define

$$X_n = \begin{cases} X & \text{with probability } 1 - \frac{1}{n} \\ e^n & \text{with probability } \frac{1}{n}. \end{cases}$$

Does X_n converge to X in probability? Does X_n converge to X in distribution? Does $\mathbb{E}(X - X_n)^2$ converge to 0?

PROOF. First we determine if X_n converges to X in quadratic mean. Observe that

$$\begin{aligned} \mathbb{E}(X_n - X)^2 &= \mathbb{E}(X - X_n)^2 = \mathbb{E}(X^2 - 2XX_n + X_n^2) \\ &= \mathbb{E}(X^2) - 2\mathbb{E}(XX_n) + \mathbb{E}(X_n^2) \end{aligned}$$

Calculating each of the expectations yields

$$\begin{aligned}
\mathbb{E}(X^2) &= (1)^2 \left(\frac{1}{2}\right) + (-1)^2 \\
\mathbb{E}(X_n^2) &= \mathbb{E}(X_n^2 \mid X = 1) \mathbb{P}(X = 1) + \mathbb{E}(X_n^2 \mid X = -1) \mathbb{P}(X = -1) \\
&= \left((1)^2 \left(1 - \frac{1}{n}\right) + e^{2n} \left(\frac{1}{n}\right)\right) \cdot \frac{1}{2} \\
&\quad + \left((-1)^2 \left(1 - \frac{1}{n}\right) + e^{2n} \left(\frac{1}{n}\right)\right) \cdot \frac{1}{2} \\
\mathbb{E}(XX_n) &= \mathbb{E}(XX_n \mid X = 1) \mathbb{P}(X = 1) + \mathbb{E}(XX_n \mid X = -1) \mathbb{P}(X = -1) \\
&= \left((1)^2 \left(1 - \frac{1}{n}\right) + (1)e^n \left(\frac{1}{n}\right)\right) \cdot \frac{1}{2} \\
&\quad + \left((-1)^2 \left(1 - \frac{1}{n}\right) + (-1)e^n \left(\frac{1}{n}\right)\right) \cdot \frac{1}{2} \\
&= 1 - \frac{1}{n} \\
\mathbb{E}(X - X_n)^2 &= 1 - 2 \left(1 - \frac{1}{n}\right) + 1 + \frac{1}{n} (e^{2n} - 1) \\
&= \frac{e^{2n}}{n}
\end{aligned}$$

Since $\lim_{n \rightarrow \infty} \mathbb{E}(X - X_n)^2 = \lim_{n \rightarrow \infty} \frac{e^{2n}}{n} \neq 0$, we have $X_n \not\stackrel{q.p.}{\rightarrow} X$.

Now we investigate if $X_n \xrightarrow{P} X$:

$$\begin{aligned}
\mathbb{P}(|X_n - X| > \epsilon) &= \mathbb{P}(0 > \epsilon \mid X_n = X) \mathbb{P}(X_n = X) \\
&\quad + \mathbb{P}(e^n - X > \epsilon \mid X_n = e^n) \mathbb{P}(X_n = e^n) \\
&= \mathbb{P}(e^n - X > \epsilon \mid X_n = e^n) \cdot \frac{1}{n}
\end{aligned}$$

Taking the limit, we get

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| > \epsilon) = \lim_{n \rightarrow \infty} \mathbb{P}(e^n - X > \epsilon \mid X_n = e^n) \cdot \frac{1}{n} = 0$$

proving that $X_n \xrightarrow{P} X$, and by Theorem 5.4(b), $X_n \rightsquigarrow X$. $\square$

Question: 5.8.10

Let $Z \sim N(0, 1)$. Let $t > 0$. Show that, for any $k > 0$,

$$\mathbb{P}(|Z| > t) \leq \frac{\mathbb{E}|Z|^k}{t^k}$$

Compare this to Mill's inequality in Chapter 4.

PROOF. By Markov's inequality,

$$\mathbb{P}(|Z| > t) = \mathbb{P}(|Z|^k > t^k) \leq \frac{\mathbb{E}|Z|^k}{t^k}$$

□

Question: 5.8.11

Suppose that $X_n \sim N(0, 1/n)$ and let X be a random variable with distribution $F(x) = 0$ if $x < 0$ and $F(x) = 1$ if $x \geq 0$. Does X_n converge to X in probability? (Prove or disprove). Does X_n converge to X in distribution? (Prove or disprove).

PROOF. We have $X \sim F$ where $F(x)$ is defined as:

$$F(x) = \begin{cases} 0 & x < 0 \\ 1 & x \geq 0 \end{cases}$$

which is the Heaviside function. Meanwhile, X_n has PDF

$$f_n(x_n) = \sqrt{\frac{n}{2\pi}} \exp\left(-\frac{x_n^2 n}{2}\right)$$

As $n \rightarrow \infty$, $\lim_{n \rightarrow \infty} f_n(x_n) = 0$ when $x_n \neq 0$. At $x_n = 0$, $\lim_{n \rightarrow \infty} f_n(x_n)$ diverges to infinity. In other words, this is the Dirac delta function:

$$\lim_{n \rightarrow \infty} f_n(x_n) = \delta(x) = \begin{cases} +\infty & x = 0 \\ 0 & x \neq 0 \end{cases}$$

Then it follows that the distribution

$$F_n(t) = \sqrt{\frac{n}{2\pi}} \int_{-\infty}^t \exp\left(-\frac{x_n^2 n}{2}\right) dx_n$$

tends toward

$$\lim_{n \rightarrow \infty} F_n(t) = \int_{-\infty}^t \delta(x) dx = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$

Thus we have

$$\lim_{n \rightarrow \infty} F_n(t) = F(t) \implies X_n \rightsquigarrow X$$

Next, since $F(x)$ is the Heaviside function, X must also have a Dirac delta PDF (there are nuances here with respect to Lebesgue integrability but that is outside the scope of this textbook). We examine if X_n converges to X in L_1 , assuming $X_n - X \geq 0$ (we will get the same result with $X_n - X < 0$ anyhow):

$$\begin{aligned} \lim_{n \rightarrow \infty} \mathbb{E}|X_n - X| &= \lim_{n \rightarrow \infty} \mathbb{E}(X_n - X) = \lim_{n \rightarrow \infty} (\mathbb{E}(X_n) - \mathbb{E}(X)) \\ &= \lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} x_n \sqrt{\frac{n}{2\pi}} \exp\left(-\frac{x_n^2 n}{2}\right) dx_n - \int_{-\infty}^{\infty} x \delta(x) dx \\ &= \int_{-\infty}^{\infty} x' \delta(x') dx' - \int_{-\infty}^{\infty} x \delta(x) dx \\ &= 0 \end{aligned}$$

This proves that $X_n \xrightarrow{L_1} X$, which implies $X_n \xrightarrow{P} X$. $\square$

Question: 5.8.12

Let $X, X_1, X_2, X_3, \dots$ be random variables that are positive and integer valued. Show that $X_n \rightsquigarrow X$ if and only if

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n = k) = \mathbb{P}(X = k)$$

for every integer k .

PROOF. ($\implies$) Let $X_n \rightsquigarrow X$. Then

$$\lim_{n \rightarrow \infty} F_n(k) = F(k)$$

where $F_n(k) = \mathbb{P}(X_n \leq k)$ and $F(k) = \mathbb{P}(X \leq k)$ for any positive integer k . Beginning from $k = 1$, we must have

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n \leq 1) = \lim_{n \rightarrow \infty} \mathbb{P}(X_n = 1) = \mathbb{P}(X \leq 1) = \mathbb{P}(X = 1)$$

Then for $k = 2$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n \leq 2) = \mathbb{P}(X \leq 2)$$

which implies that $\lim_{n \rightarrow \infty} \mathbb{P}(X_n = 2) = \mathbb{P}(X = 2)$, as we ascertained that $\lim_{n \rightarrow \infty} \mathbb{P}(X_n = 1) = \mathbb{P}(X = 1)$. We can continue in this fashion to conclude

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n = k) = \mathbb{P}(X = k)$$

($\impliedby$) Let $\lim_{n \rightarrow \infty} \mathbb{P}(X_n = k) = \mathbb{P}(X = k)$ for every positive integer k . If we observe that

$$\begin{aligned} \lim_{n \rightarrow \infty} \mathbb{P}(X_n \leq k) &= \lim_{n \rightarrow \infty} \sum_{i=1}^k \mathbb{P}(X_n = i) \\ \mathbb{P}(X \leq k) &= \sum_{i=1}^k \mathbb{P}(X = i) \end{aligned}$$

Then we can conclude

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n \leq k) = \mathbb{P}(X \leq k)$$

which is equivalent to claiming

$$\lim_{n \rightarrow \infty} F_n(k) = F(k) \implies X_n \rightsquigarrow X$$

□

Question: 5.8.13

Let $Z_1, Z_2, \dots$ be IID random variables with density f . Suppose that $\mathbb{P}(Z_i > 0) = 1$ and that $\lambda = \lim_{x \downarrow 0} f(x) > 0$. Let

$$X_n = n \min\{Z_1, \dots, Z_n\}.$$

Show that $X_n \rightsquigarrow Z$ where Z has an exponential distribution with mean $1/\lambda$.

PROOF. First, recall that the CDF of $Z \sim \exp(1/\lambda)$ is

$$F(t) = \int_0^t \lambda \exp(-\lambda x) dx = 1 - \exp(-\lambda t)$$

Here's our intuition: does $\mathbb{P}(X_n \leq t)$ follow a pattern that allows us to derive an exponential function by way of a limit?

For $X_n \leq t$, we must have $\min\{Z_1, \dots, Z_n\} \leq t/n$. To guarantee this is the case, we must guarantee that *all* the $Z_i \leq t/n$ for all i . Then since the random variables are IID, it follows that

$$\begin{aligned} \mathbb{P}(X_n \leq t) &= \prod_{i=1}^n \mathbb{P}\left(Z_i \leq \frac{t}{n}\right) \\ &= 1 - \prod_{i=1}^n \mathbb{P}\left(Z_i \geq \frac{t}{n}\right) \\ &= 1 - \prod_{i=1}^n \left(\mathbb{P}(Z_i > 0) - \mathbb{P}\left(Z_i \leq \frac{t}{n}\right)\right) \\ &= 1 - \prod_{i=1}^n \left(1 - \mathbb{P}\left(Z_i \leq \frac{t}{n}\right)\right) \end{aligned}$$

Now, for $n \rightarrow \infty$, for fixed t , $t/n \rightarrow 0$. By premise, $\lim_{x \downarrow 0} f(x) = \lambda$. For infinitesimal t/n , we can approximate $\mathbb{P}(Z_i \leq t/n) = \int_0^{t/n} f dz_i$ as follows:

$$\lim_{n \rightarrow \infty} \mathbb{P}\left(Z_i \leq \frac{t}{n}\right) = \lim_{n \rightarrow \infty} \int_0^{t/n} f dz_i = \frac{\lambda t}{n}$$

where we can think of $f \approx \lambda$ and $\Delta z_i \approx t/n$. Now we can finish with

$$\begin{aligned}
 \lim_{n \rightarrow \infty} F_n(t) &= \lim_{n \rightarrow \infty} \mathbb{P}(X_n \leq t) \\
 &= 1 - \lim_{n \rightarrow \infty} \prod_{i=1}^n \left(1 - \mathbb{P}\left(Z_i \leq \frac{t}{n}\right)\right) \\
 &= 1 - \lim_{n \rightarrow \infty} \prod_{i=1}^n \left(1 - \frac{\lambda t}{n}\right) \\
 &= 1 - \lim_{n \rightarrow \infty} \left(1 - \frac{\lambda t}{n}\right)^n \\
 &= 1 - \exp(-\lambda t) \\
 &= F(t)
 \end{aligned}$$

□

Question: 5.8.14

Let $X_1, \dots, X_n \sim \text{Uniform}(0, 1)$. Let $Y_n = \overline{X}_n^2$. Find the limiting distribution of Y_n .

Recall that for $X_1 \sim \text{Uniform}(0, 1)$, $\mathbb{E}(X_1) = 1/2$ and $\mathbb{V}(X_1) = 1/12$. Then $\mathbb{E}(\overline{X}_n) = 1/2$ and $\mathbb{V}(\overline{X}_n) = 1/12n$. By Theorem 5.13 (the Delta Method), since $\overline{X}_n \approx N(1/2, 1/12n)$, we have

$$Y_n = \overline{X}_n^2 \approx N\left(\frac{1}{4}, \frac{1}{12n}\right)$$

Question: 5.8.15

Let

$$\begin{pmatrix} X_{11} \\ X_{21} \end{pmatrix}, \begin{pmatrix} X_{12} \\ X_{22} \end{pmatrix}, \dots, \begin{pmatrix} X_{1n} \\ X_{2n} \end{pmatrix}$$

be IID random vectors with mean $\mu = (\mu_1, \mu_2)$ and variance Σ . Let

$$\overline{X}_1 = \frac{1}{n} \sum_{i=1}^n X_{1i}, \quad \overline{X}_2 = \frac{1}{n} \sum_{i=1}^n X_{2i}$$

and define $Y_n = \overline{X}_1 / \overline{X}_2$. Find the limiting distribution of Y_n .

Let $Y_n = g(\overline{X}_1, \overline{X}_2)$ where $g(s_1, s_2) = s_1/s_2$. The gradient is

$$\nabla g(s) = \begin{pmatrix} \frac{\partial g}{\partial s_1} \\ \frac{\partial g}{\partial s_2} \end{pmatrix} = \begin{pmatrix} \frac{1}{s_2} \\ -\frac{s_1}{s_2^2} \end{pmatrix}$$

Then by the multivariate Delta Method,

$$\begin{aligned}\nabla_{\mu}^T \Sigma \nabla_{\mu} &= \begin{pmatrix} 1/\mu_2 & -\mu_1/\mu_2^2 \end{pmatrix} \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{pmatrix} \begin{pmatrix} 1/\mu_2 \\ -\mu_1/\mu_2^2 \end{pmatrix} \\ &= \begin{pmatrix} \frac{\sigma_{11}}{\mu_2} - \frac{\mu_1 \sigma_{12}}{\mu_2^2} & \frac{\sigma_{12}}{\mu_2} - \frac{\mu_1 \sigma_{22}}{\mu_2^2} \end{pmatrix} \begin{pmatrix} 1/\mu_2 \\ -\mu_1/\mu_2^2 \end{pmatrix} \\ &= \frac{\sigma_{11}}{\mu_2^2} - \frac{2\mu_1 \sigma_{12}}{\mu_2^3} + \frac{\mu_1^2 \sigma_{22}}{\mu_2^4}\end{aligned}$$

Thus we can conclude

$$\sqrt{n} \left(\frac{\bar{X}_1}{\bar{X}_2} - \frac{\mu_1}{\mu_2} \right) \rightsquigarrow N \left(0, \frac{\sigma_{11}}{\mu_2^2} - \frac{2\mu_1 \sigma_{12}}{\mu_2^3} + \frac{\mu_1^2 \sigma_{22}}{\mu_2^4} \right)$$

Question: 5.8.16

Construct an example where $X_n \rightsquigarrow X$ and $Y_n \rightsquigarrow Y$ but $X_n + Y_n$ does not converge in distribution to $X + Y$.

Define $X, Y \sim N(0, 1)$, $X_n = X$, $Y_n = -X$, $Z = X + Y$, and $Z_n = X_n + Y_n$. It follows that $Z \sim N(0, 2)$. Moreover, we satisfy the premise that $X_n \rightsquigarrow X$ and $Y_n \rightsquigarrow Y$; it is trivial in the former case, and for the latter, simply observe that

$$f_X(-x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(-x)^2}{2}\right) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) = f_X(x)$$

Since $Z_n = X_n + Y_n = 0$ is a constant random variable, its distribution is

$$F_n(Z_n) = \begin{cases} 0 & Z_n < 0 \\ 1 & Z_n \geq 0 \end{cases}$$

demonstrating that $Z_n \not\rightsquigarrow Z$.